Balb/C myeloma cells that stick to flask bottom can be resuspended easily by gentle scraping with a plastic Pasteur pipette without trypsin?

a. False

b. True

Chemically defined medium for high density cell culture have no serum proteins that might contaminate Mabs

O a. True

Ob. False

The human Mabs are more useful than murine Mabs for in vitro diagnostic purposes

a. False

b. True

The following applies for determination of Mab titer using ELISA method?

a. The titer is defines as the lowest dilution of Ag to bind significantly to Mab

b. The titer is defines as the highest dilution of Mab to bind significantly to Ag

c. Serial dilutions of a Mab are allowed to react with known constant amount of Ag

d. Serial dilutions of the Ag are allowed to react with known constant amount of Mab

The following characterize HAT media used in Mab production?

a. A stands for Aminopterin as inhibitor of De novo DNA synthesis

b. T stands for thymidine as inhibitor of salvage DNA synthesis pathway

c. T stands for thymol that inhibit both pathways of DNA synthesis

d. H stands for Hypoxanthin as inhibitor of DNA salvage synthesis

The following does not apply for propagation of monoclonal antibodies in vitro

a. prior to hybridoma cell injection, the mice must be injected with prista

O b. The mice must be adult (3-6 months)

O c. Ascitic fluid produced is rich in the Mabs

d. the injected mice can be of any strain other than Balb/c.

The following is false about production of Mabs by hollow fiber bioreactor?

a. Passaging of cells is not required

b. Enrichment of TGF beta factor

c. relatively small size

d. High yield with high titer

The following applies for the affinity of a Mab?

a. High affinity Mab is preferable for use in immunoassays

b. Affinity reflects the concentration of a Mab in immunoassay

c. Affinity is the description of the titer of a Mab

d. Affinity reflects the specificity of a Mab in immunoassays

The following does not characterize polyethylene glycol (PEG) used in hybridoma production? A

a. high fusion rate

b. Molecular weight 1500-4000

c. high cytotoxicity

d. concentration 40-50%

The following results are obtained with SDS-PAGE analysis to ensure purity of a Mab?

O a. 3 bands

b. 2 bands

c. 1 band

d. 4 bands

The following is false about limiting dilution for cloning of Ag-specific hybridoma

a. Negative wells are expanded and recloned at least 3 times .

b. positive wells are expanded and recloned at least 3 times

c. The 1 cell/well clone should display positive Mab in all seeded wells

d. before cloning, the hybridoma culture should be tested for the presence of the desired Mab

The following is correct about feeder cells used in hybridoma cultures?

a. are liver cells that are rich in macrophages

b. is necessary to use the same species (Balb/c) as the hybridoma parents

c. are lymph nodes taken from young adult animals

d. Peritoneal macrophages are usually used

The following is not correct for long storage of myeloma cells?

a. the storage media contains formalin in RPMI for sterility.

b. using cryotubes

c. Stored in liquid nitrogen

d. Cells should be placed first in polystyrene box at -80 OC

In hollow fiber bioreactor during production of Mabs?

a. The extra capillary tubes filled with sterile media

b. If the expected levels of Mab in culture are diminishing, the hybridoma cells should b

c. The fidelity of the specific Mab production should be monitored regularly due to out

d. the capillary tubes are filled with the hybridoma cells

The following is true about selection of hybridoma after fusion in HAT medium

a. Unfused B cells die within 10-14-days.

b. The Ag-Specific and nonspecific hybridoma cells survive

c. Fused myeloma -myeloma and B-B cells die within 10-14 days

d. Unfused Myeloma cells die within 10-14 days

The following method is not used in the purification of Mabs?

a. Protein A affinity chromatography

b. Gel filtration according to size

c. Ammonium sulfate precipitation

d. SDS-polyacrylamide gel electrophoresis

The following applies to fusion protocol between B and myeloma cells?

a. Cells are incubated in HAT medium for 10-14 days

b. 5% FCS media is recommended

c. After addition of PEG, the cells should be incubated in HAT media immediately

d. PEG is added once in one shot very quickly to avoid toxicity

The following solvents or detergents are not used in the viral inactivation of Mabs?

a. Ammonium sulfate

b. TNPB and sodium cholate

c. Triton-100

d. Caprylic acid

The following applies for affinity determination of a Mab by inhibition radioimmunoassay.

a. Serial dilutions of unlabeled Abs are dispensed into assay tubes

b. Scatchard plot is applied for affinity measurement.

c. Constant amounts of unlabelled Abs are added to each tube.

d. Constant amount of Ag is added to all tubes.

The following describe humanized Mab

a. the constant region of a Mab and the framework regions are of human origin

b. In addition to the constant region of a Mab, the CDRs regions are replaced by human

c. The variable region is replaced by human while the constant region remain murine

d. The constant region is replaced by human while the variable region remain murine

The following does not characterize fusion conditions between B and myeloma cells?

a. The ratio is 1.10: 1 of Myeloma: B cells

b. Optimal is room temperature

c. Concentration of PEG 40-50%

d. Polyethylene glycol is used

The following describe chimeric Mab?

a. The mouse constant region and the CDRS regions of a Mab, are replaced by human

b. The constant region is replaced by human while the variable region remain murina

c. The constant region of a Mab and the framework regions are of mouse origin

d. The variable region is replaced by human while the constant region remain murina

The following applies for the affinity of a Mab?

a. Low affinity is essential for use in immunoassays

b. Affinity is the description of the specificity between a Mab and an Ag

c. High affinity is preferable for use in immunoassays

d. High affinity is preferable for use in immunepurification

The recommended adjuvant for in vitro immunization is?

a . Complete Freund adjuvant

b. incomplete Freund adjuvant

c. CPG

d. Muramyl dipeptide

To prove specificity of a Mab?

a. It should react with strongly affinity with the specific Ag

b. It should be immunologically different from other Mabs.

c. It should be tested whether it reacts with related Ags or not.

d. It must have a high titer against the specific Ag

Prior to fusion between B cells and myeloma cells, it is essential that?

a. cell viability should be more than 59%

b. Cell density should be not more than 10 x 106 cells/ml

c. B cells should be pure

d. The myeloma cells should be in log phase

Thymus cells as feeder layer during hybridoma culturing?

a. provide growth factors

b. should be obtained from young animal

c. It is not necessary to use the same species (Balb/c) as the hybridoma parents

d. All choices are correct

After fusion experiment, the purpose of cloning method is?

a. isolation of Ag specific Mabs

b. isolation of single Ag specific B cell

c. isolation of Ag specific myeloma cell

d. isolation of Ag specific hybridoma cell

For separation of Ag specific from nonspecific human B cells, the following positive selection method is effective?

a. ELISA using specific Abs

b. Magnetic beads coated with Abs against CD19

c. Ficoll-hypaque separation method

d. Ammonium sulfate precipitation method.

Isotype determination of a monoclonal Ab means?

Oa. Determination of its titer

Ob. Determination of its class and subclass

Oc. Determination of its binding strength

Od. Determination of its specificity

The mouse spleen is chosen as a source of B lymphocytes because of the following?

a. rich in several immune cells including monocytes and macrophages

b. has connective tissue that makes cell release easily

c. rich in B cells

d. Has large size

To test whether the separated hybridoma clones are identical or totally different, the following test could be performed?

a. specific agglutination test

b. specific immunoelectrophoresis

c. especific double Immunodiffusion

d. Competitive inhibition radioimmunoassay

To test the effectiveness of HAT, myeloma cells are cultured in HAT media. The following results should be obtained

a. Myetoma celts the

b. myeloma cells survive

c. Charmy choice

it is inadvisable to maintain myeloma cells in culture indefinitely since?

a. they become mortal

b. They lose the proliferation property

c. They lose the proliferation property.

d. revert to HOPRT+

It is inadvisable to maintain myeloma cells in culture indefinitely since?

a. they become mortal

b. They lose Ab secretion

c. They might revert to HGPRT+

d. They lose the proliferation property

Fetal calf serum used in tissue culture to support the growth of B cells?

a. is rich in IgG to prevent contamination

b. is IgG depleted by using protein G affinity chromatography

To test the effectiveness of HAT, myeloma cells or hybridoma cells should grow in HAT media. The following results should be obtained?

a. Both cells survive

b. Hybridoma cells dio while myeloma cells survive

b. Hybridoma cells die while myeloma cells survive

c. Both oils die

d. myeloma cells die while hybridoma survive

Selection of HGPRT- myeloma cells is carried out in the presence of?

a. formaldehyde

b. Thymidine

c. cytidine

d. 8-azaguanine

Why human spleen or tonsils make better fusion partner with myeloma than do from blood?

a. In blood, few antigen presenting cells (APCS)

b. All choices are correct

c. in blood, not enough Ag-specific B cells in a state of proliferation and differentiation

d. In blood, more suppressor cells

Usually at high cell dilutions, the growth of hybridomas is inhibited. This can be overc

O a. concentration of the hybridoma cells by evaporation of water in the incubator.

O b. addition of BSA to increase the tonicity and concerration

O c. decrease of PH by acidic buffers

d. Addition of feeder cell layer

?Isolated Immune cells from mouse peritoneum include

a. Dendritic cells and Mast cells

b. immature neurons

c. epithelial cells

d. Macrophages and B cells

Expansion of hybridoma cells prior to cloning is performed by?

O a. 96-well plates used instead of 8 well plates.

Ob. absence of FCS or feeder layer cells

Oc. HT medium is replaced by HAT

d. HAT medium is replaced by HT medium

According to density gradient centrifugation of human blood using ficoll-hypaque, the buffy coat contain the following cells?

a. lymphocytes and platelets

b. granulocytes and lymphocytes

c. Plasma and ficoll

d. monocytes and RBCs

In phage display method for production of human Mab, the following step is not included?

a. A gene library of this phage displaye lg can be made and Selection of the phage that produce the Mab

b. Isolation of mRNA then converting it into cDNA by PCR to amplify the lg variable region

c. Isolation of myeloma cells from the blood of the human body

d. The lg genes are inserted into phage directly.

For ascitic production, injection the mouse with high number of hybridoma cells is not recommended because?

a. feedback inhibition of cell proliferation

b. sudden unexpected mice death might occur

Oc. Larger volume of ascitic fluid usually produced.

Od. weak affinity Mabs are produced

The continued storage of cells at all stages of Mab production is essential because?

a. There is always a risk of overgrowth by Ig-nonproducing hybridomas

b. All choices are correct

c. Throughout cloning, there is a high probability of chromosome loss and consequent loss of lg secretion

d. The risk of infection

The requirement for continued storage of cell at all stages of Mao pr

a. Throughout cloning, there is a high probability of chromosome loss and consu secretion

b The Wok of infection

c. there is always a risk of overgrowth by ig- nonproducing hybridomas

d. all choices are correct

To count myeloma cells, trypan blue exclusion test is used. The disadvantage of trypan blue?

a. Stain mainly the viable cells

b. Have great affinity to serum proteins

c. Difficult to see cells under the microscope

d. Toxic to viable cells

By EBV transformation, the following human Mabs class could be produced from any individual

a. IgG only

b. IgA Only

c. Any lg class

d. IgM Only

Normal cells usually divide only a limited number of times before losing their ability to proliferate, wi genetically determined event known as tonoscence. Those collines are called?

a Finite

b. Cancera

Hybridoma cells have an application to produce :

a. Antibodies

b. Cancer cells

c. Antigens

d. Cell lines

Human myelomas are difficult to establish in tissue culture because all of the following EXCEPT?

a. doubling time in culture is about 70 hours

b. are aneuploid

c. have abundant RER and mitochondria with high lg production

d. are large size

The preparation of single cell suspension from mouse spleen include?

a. smashing the spleen between 2 frosted slides

b. using 3 different syringes needles (18, 22, and 26 guage)

c. All choices are correct

Single cell suspension from mouse spleen could be prepared by?

O a. Specific antibodies are added for cell disruption

O b. treatment with trypsin for cell separation

O c. cut spleen into 3-4 pieces followed by washing with media to release

Od. using mesh strainer after the spleen is minced